

Grade 2 Ontario Curriculum Spatial Sense

Measure Maven

Unit Overview: Measure Maven

Grade: Grade 2 | **Strand:** Spatial Sense

Target Expectations: E2.1, E2.2, E2.3, E2.4

Learning Goal: Children will choose units, use centimetres/metres + benchmarks, measure with rulers, recognize zero-point, and describe duration using time units.

Lesson	Lesson Title	Learning Goal
Day 1	Pick Your Unit	I can pick a non-standard unit and explain the inverse relationship.
Day 2	Centimetres and Metres	I can use cm and m and estimate with benchmarks.
Day 3	Measure with Rulers	I can measure and draw lengths with a ruler.
Day 4	How Long Did That Take?	I can describe duration using the right time unit.
Day 5	Measure Marathon Capstone	I can measure length and time using the right tool.

Lesson 1: Pick Your Unit

Goal: Smaller units = more units needed. Bigger units = fewer units.
INVERSE relationship.

Expectation: E2.1

Learning Goal: "I can pick a non-standard unit and explain the inverse relationship."

Materials: M1_unit_set, M10_unit_poster

1. Minds On (12 mins): Roo Measures Two Ways

Roo: A pencil = 12 CUBES long! But only 2 HAND-SPANS! Smaller units = MORE.
Bigger units = FEWER! Roo opens: 'SMALL unit, BIG count — BIG unit, small count!'

Bring children to the carpet.

2. Action (25 mins): Unit Stations

- Whole group: measure 2 objects with cubes AND hand-spans.
- Three stations.
- Stretch at C: measure same distance 3 ways and compare.
- Discuss: smaller unit, bigger count.
- Circulate to record diagnostic evidence.
- PREDICT-THEN-MEASURE: Child PREDICTS the count for a hand-span before measuring, then compares with a partner who used cubes.

3. Consolidation (13 mins): Roo's Unit Brag

Discuss:

- What's a NON-STANDARD UNIT?
- Why do smaller units give bigger counts?
- When use cubes?
- What's the INVERSE relationship?
- Roo's preview: tomorrow STANDARD units (cm, m) so EVERY classroom counts the same way (Lesson 2 — E2.1).

Exit: Roo's refrain — chant 3x: 'SMALL unit, BIG count — BIG unit, small count!'
Each child records one object with two units.

Lesson 2: Centimetres and Metres

Goal: 100 cm = 1 m. Use BENCHMARKS — fingernail = 1 cm, hockey stick = 1 m.

Expectation: E2.2

Learning Goal: "I can use cm and m and estimate with benchmarks."

Materials: M2_cm_strips, M3_metre_strip, M7_benchmark_chart, M10_unit_poster

1. Minds On (12 mins): Roo Introduces Standard Units

Roo: STANDARD UNITS! Centimetres are SMALL. Metres are BIG. 100 centimetres = 1 metre! Roo opens with the unit refrain: 'SMALL unit, BIG count — BIG unit, small count!'

Yesterday Roo measured with HAND-SPANS — but Aaron's hand-span and Bea's hand-span are different sizes! Today STANDARD units (cm, m) make sure EVERY classroom counts the same way.

2. Action (25 mins): cm and m Stations

- Whole group: estimate 4 lengths with benchmarks.
- Three stations.
- Stretch at C: estimate 5 different lengths in mixed cm/m.
- Discuss when to use cm vs m.
- Circulate to record formative evidence.
- STRATEGY COMPARE: Two pairs share HOW they estimated the doorway — whose-arm-span vs hockey-stick-count vs counting tiles.

3. Consolidation (13 mins): Roo's cm/m Brag

Discuss:

- What's a BENCHMARK?
- How many cm in a metre?
- What's a 1-cm benchmark?
- When use cm?

Exit: Roo's standard refrain — 'SMALL unit, BIG count — BIG unit, small count!' (3x).

Lesson 3: Measure with Rulers

Goal: Always START at ZERO on the ruler. Read the number at the END of the object.

Expectation: E2.3

Learning Goal: "I can measure and draw lengths with a ruler."

Materials: M2_cm_strips, M3_metre_strip, M4_ruler_set, M8_zero_chart

1. Minds On (12 mins): Roo Measures with a Ruler

Roo: My ruler starts at ZERO! Pencil end touches ZERO. Other end shows TEN. Pencil = 10 cm! Roo opens: 'COUNT the SPACES, start at ZERO, end at the EDGE!'

Bring children to the carpet. Roo and Cal puppets. 'Friends — yesterday cm and m. Today RULERS.'

2. Action (25 mins): Ruler Stations

- Whole group: measure 4 objects with rulers.
- Three stations.
- Stretch at C: spot the error in 3 wrong-zero examples.
- Discuss zero-point.
- Circulate to record formative evidence.
- STRATEGY COMPARE: Two children measure the same crayon — one starting at 0, one starting at 2.

3. Consolidation (13 mins): Roo's Ruler Brag

Discuss:

- Where do we START on a ruler?
- What's the rule?
- What if I start at 1?
- How do we DRAW a 5-cm line?

Exit: Roo's ruler refrain — 'COUNT the SPACES, start at ZERO, end at the EDGE!' (3x).

Lesson 4: How Long Did That Take?

Goal: TIME = SECONDS, MINUTES, HOURS. We pick the right unit for the duration.

Expectation: E2.4

Learning Goal: "I can describe duration using the right time unit."

Materials: M5_time_chart, M9_event_cards, M10_unit_poster

1. Minds On (12 mins): Cal Times Events

Cal: A BLINK = 1 SECOND. Brushing teeth = 2 MINUTES. School day = 6 HOURS!
Cal opens: 'SECONDS for the FAST, MINUTES for the MEDIUM, HOURS for the LONG!'

Remember Day 3: the ruler told us LENGTH in cm.

2. Action (25 mins): Time Stations

- Whole group: pick units for 4 events.
- Three stations.
- Stretch at C: estimate before timing.
- Discuss the 60 rule.
- Circulate to record formative evidence.
- STRATEGY COMPARE: After Station B, two pairs share HOW they ESTIMATED before timing — Pair A 'Mississippi-counting'; Pair B 'body-pose method'.

3. Consolidation (13 mins): Cal's Time Brag

Discuss:

- What's a SECOND?
- How many seconds in a minute?
- How long is recess?
- How long is a day?
- Cal's preview: tomorrow Roo + Cal join forces — measure ALL attributes (length, mass, capacity, time) at the capstone stations...

Exit: Cal's time refrain — 'SECONDS for the FAST, MINUTES for the MEDIUM, HOURS for the LONG!' (3x).

Lesson 5: Measure Marathon Capstone

Goal: We measure LENGTH and TIME with the right TOOLS and UNITS.

Expectation: E2.1, E2.2, E2.3, E2.4

Learning Goal: "I can measure length and time using the right tool."

Materials: M1_unit_set, M2_cm_strips, M3_metre_strip, M4_ruler_set

1. Minds On (12 mins): Roo and Cal's Marathon Plan

Roo: Today's MARATHON — 4 stations: pick a unit, use cm/m, ruler, time! Be the MEASURE MAVEN! Roo + Cal open: 'SMALL unit, BIG count — BIG unit, small count! Right TOOL, right NUMBER!'

Bring children to the carpet. Roo and Cal puppets in lab coats. 'Friends — today is our CAPSTONE. We use ALL: non-standard units, cm/m, ruler, time.'

2. Action (25 mins): The Measure Marathon

- Hand out templates.
- Station 1 (UNIT): pick + measure 3 objects.
- Station 2 (cm/m): estimate 4 lengths.
- Station 3 (RULER): measure + draw.
- Station 4 (TIME): pick units for 4 events.
- STRATEGY COMPARE at Station 3 (RULER): two children measure the same pencil — one starting at 0, one starting at 2.

3. Consolidation (13 mins): Roo and Cal's Marathon Brag

Discuss:

- What unit for a desk?
- How many cm in a metre?
- Where do we start on a ruler?
- How long is recess?

Exit: Closing capstone refrain — Roo + Cal call: 'SMALL unit, BIG count!' Class: 'BIG unit, small count!' Cal: 'Right TOOL, right NUMBER!' (3x).

Date: _____

Learning Goal: I can pick a non-standard unit and explain the inverse relationship.

1. Match each object to the BEST non-standard unit (cubes/hand-spans/footsteps).
2. Measure a pencil with cubes AND hand-spans.
3. Pick 3 home objects.

Match each object to the BEST non-standard unit (cubes/hand-spans/footsteps).

[illegible]

Date: _____

Learning Goal: I can use cm and m and estimate with benchmarks.

1. Estimate 5 lengths using benchmarks.
2. Convert: $100\text{ cm} = ?\text{ m}$.
3. Find 3 things at home: 1 about 1 cm, 1 about 50 cm, 1 about 1 m.

Convert: 100 cm = ? m. 200 cm = ? m. 1 m = ? cm. 2 m = ? cm.
50 cm = ? m.

[illegible]

Name: _____

Date: _____

Roo's Ruler Workout

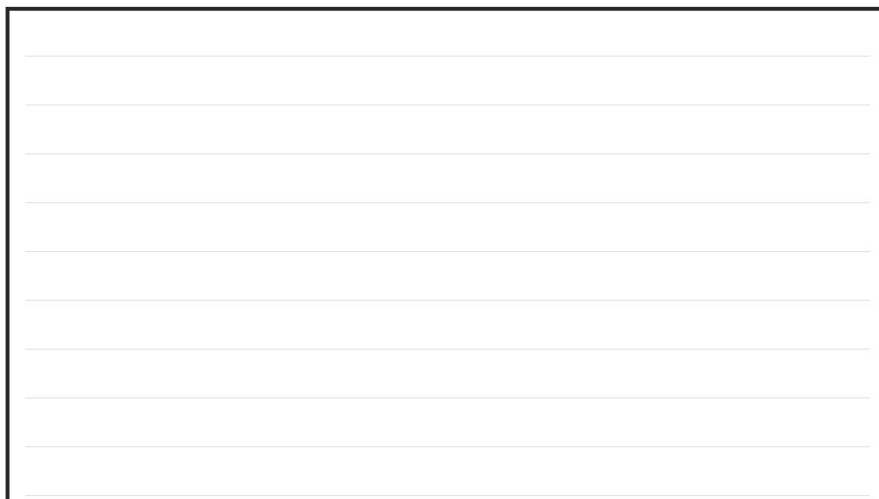
Learning Goal: I can measure and draw lengths with a ruler.

Instructions for the Student:

1. Measure 5 objects in cm.
2. Draw 4 lines: 3 cm, 5 cm, 8 cm, 12 cm.
3. Look at 2 wrong measurements.

Part 1: Measure 5 objects.

Measure 5 objects in cm. Use the ruler. START AT ZERO!



Name: _____ Date: _____

Name: _____ Date: _____

Cal's Time Test

Learning Goal: I can describe duration using the right time unit.

Instructions for the Student:

1. For each event, circle SECONDS, MINUTES, or HOURS.
2. Convert: 60 seconds = ? min.
3. List 3 events you do at home and pick units.

Part 2: Convert.

Convert: 60 seconds = ? min. 2 hours = ? min. 1 hour = ? sec
(challenge!).

[illegible]

Name: _____

Date: _____

Roo and Cal's Marathon

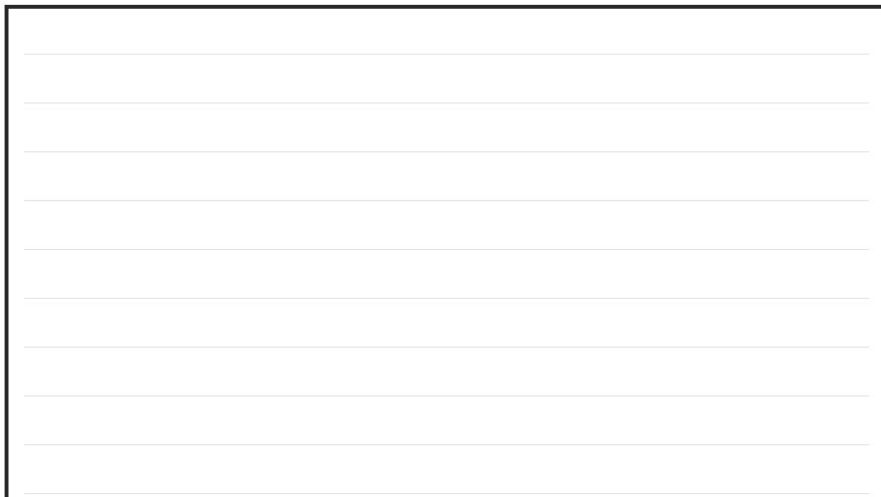
Learning Goal: I can measure length and time using the right tool.

Instructions for the Student:

1. Match 4 objects to the right unit (cubes/hand-spans/cm/m).
2. Measure 3 objects with a ruler.
3. Pick the right time unit for 4 events.

Part 2: Use the ruler.

Measure 3 objects with a ruler. Draw a 6-cm line.

A large rectangular box with a black border, containing ten horizontal grey lines for drawing and writing.

Non-Standard Unit Set

(M1_unit_set)

60 paper cubes + 12 hand-spans + 12 footprint cards.

Benchmarks

1 cm fingernail	10 cm thumb to pinky
1 m long step	1 g paper clip
100 g stick of butter	1 kg 1 L of water
1 mL drop	1 L milk carton

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 3

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into individual pieces.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 5

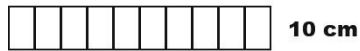
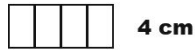
Prep: ~35 min

Centimetre Strips

(M2_cm_strips)

24 paper strips with cm markings.

Centimetre Strips



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 24 strips per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 2, 3, 5

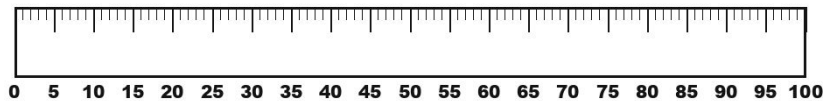
Prep: ~20 min

Metre Strip

(M3_metre_strip)

Letter-size 1-metre strip with 100 cm markings.

Ruler — 100 cm



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 10-12 metre strips

Lessons: 2, 3, 5

Prep: ~25 min

TEACHER PREP

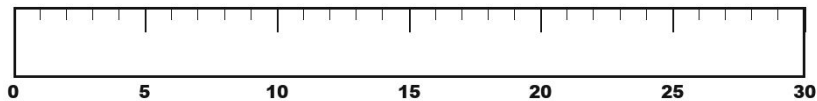
1. Print 12 sets (1 per pair).
2. Tape 2 pages together to make 1 m.
3. Laminate.
4. Distribute one per pair.

Paper Ruler Set

(M4_ruler_set)

12 paper rulers (15 cm each).

Ruler — 30 cm



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 12 rulers per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 3, 5

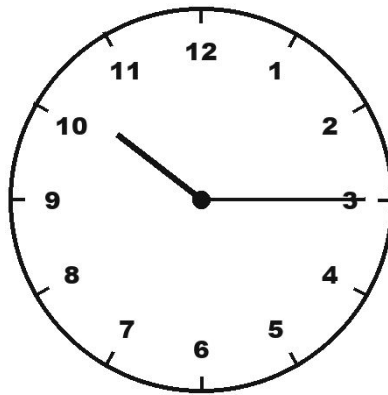
Prep: ~20 min

Time Chart

(M5_time_chart)

Letter-size chart of seconds/minutes/hours examples.

Clock — 10:15



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 4.

USE IN CLASS

Quantity: 1 chart

Lessons: 4, 5

Prep: ~10 min

Measure Marathon Template

(M6_capstone_template)

Letter-size capstone with 4 stations.

Measure Marathon Template	
1	Pick Unit _____
2	cm/m ESTIMATION _____
3	Ruler _____
4	Time _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 25 templates.

2. Stack near teacher table.

3. No lamination.

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

Benchmark Chart

(M7_benchmark_chart)

Anchor chart of cm/m benchmarks.

Benchmarks

1 cm fingernail	10 cm thumb to pinky
1 m long step	1 g paper clip
100 g stick of butter	1 kg 1 L of water
1 mL drop	1 L milk carton

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 2.

USE IN CLASS

Quantity: 1 chart

Lessons: 2, 5

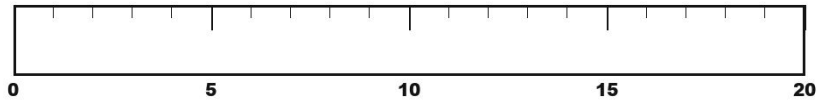
Prep: ~10 min

Start at Zero Chart

(M8_zero_chart)

Anchor chart explaining ruler zero-point.

Ruler — 20 cm



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape next to ruler chart.

USE IN CLASS

Quantity: 1 chart

Lessons: 3, 5

Prep: ~10 min

Time Event Cards

(M9_event_cards)

16 cards with everyday events.

Calendar — June

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 16 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

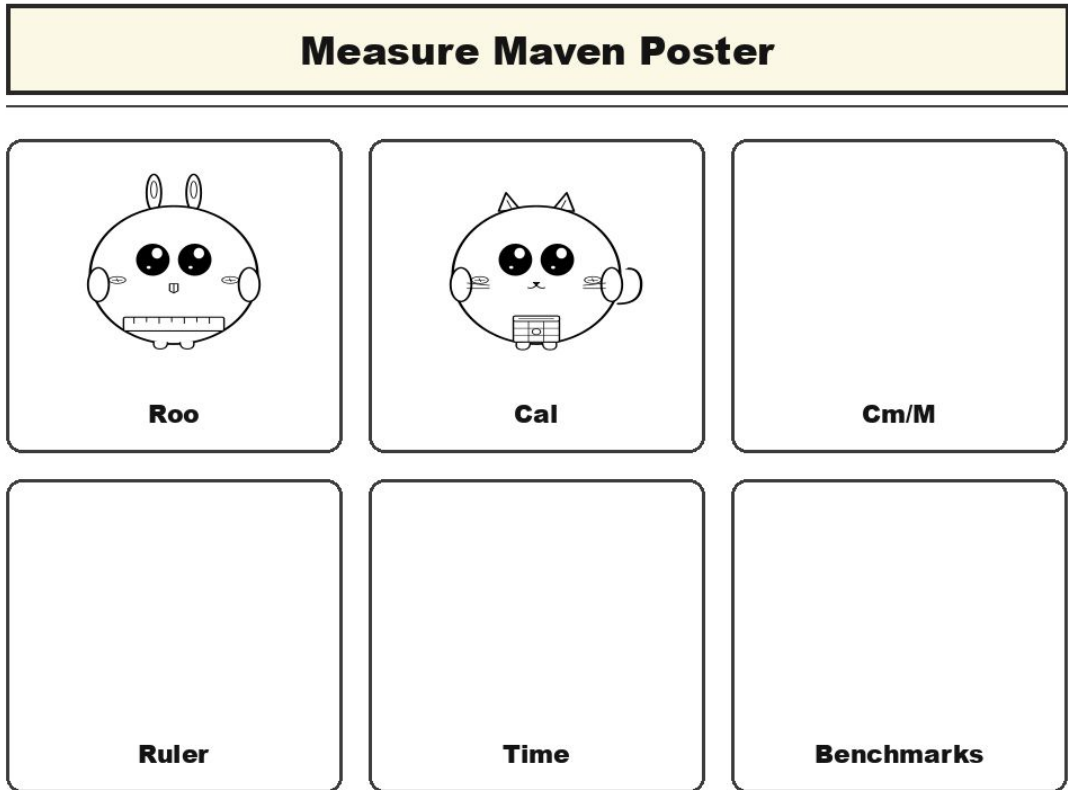
Lessons: 4, 5

Prep: ~15 min

Measure Maven Poster

(M10_unit_poster)

Unit anchor poster of Roo and Cal.



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 copy on letter or 11x17.
2. Laminate.
3. Tape to front wall on Day 1.

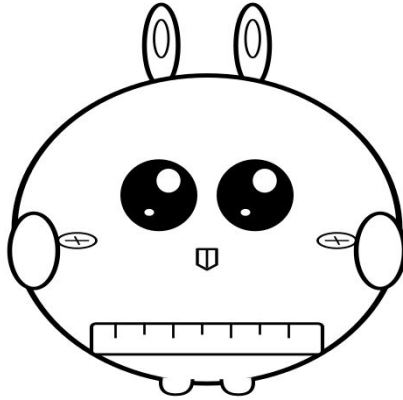
USE IN CLASS

Quantity: 1 poster
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

Roo Puppet

(char_roo_puppet)

Paper puppet of Roo for teacher use in all 5 lessons.



Roo the Ruler Rabbit

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

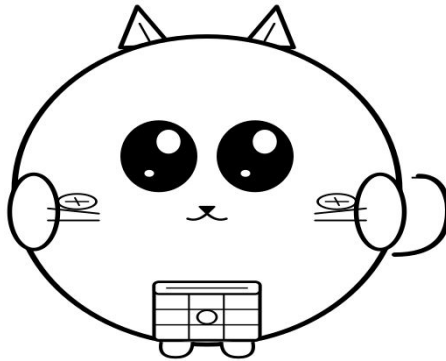
Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

Cal Puppet

(char_cal_puppet)

Paper puppet of Cal for teacher use in Lessons 4-5.



Cal the Calendar Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

Lessons: 4, 5

Prep: ~12 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
E2.1 choose and use non-standard units appropriately to measure lengths, and describe the inverse relationship between the size of a unit and the number of units	Uses cubes for everything. Cannot articulate inverse.	Picks unit with prompts.	Independently picks the right unit and articulates inverse relationship.	Predicts count from unit choice, demonstrates inverse with multiple units.
E2.2 explain the relationship between centimetres and metres as units of length, and use benchmarks for these units to estimate lengths	Doesn't know $100\text{ cm} = 1\text{ m}$.	Recalls relationship with prompts.	Independently uses cm/m + benchmarks for estimation.	Estimates within 10% accuracy and converts mixed mm/cm/m .
E2.3 measure and draw lengths in centimetres and metres, using a measuring tool, and recognize the impact of starting at points other than zero	Off-by-one (starts at 1).	Starts at zero with prompts.	Independently starts at zero and draws lengths accurately.	Recognizes and corrects zero-errors in others' work.
E2.4 use units of time, including seconds, minutes, hours	Wrong magnitude (brushing teeth = seconds).	Picks right unit with prompts.	Independently picks right unit and uses 60 rule.	Estimates duration accurately and reasons about magnitude.

Junior Measure Maven Certificate (G2)

This certificate is proudly presented to:

For successfully completing the 5-Day Measure Maven unit and demonstrating the skills of a G2 Junior Measure Maven. Roo the Ruler Rabbit congratulates the student.

By completing this unit, this student has demonstrated:

- Picking the right non-standard unit
- Using cm/m with benchmarks
- Measuring with a ruler from zero
- Picking time units for events

Marketplace Listing

A 5-day Grade 2 unit on standard measurement framed around Roo the Ruler Rabbit and Cal the Calendar Cat.

Price: \$8.99 CAD

Pages: ~42

Classroom time: 250 min

Teacher prep: ~200 min

Top tags: Grade 2, math, measurement, centimetre, metre, ruler, time, duration

What's included:

- Unit blueprint (1 file)
- 5 lesson plans (5 files)
- 5 worksheets (5 files)
- Manipulatives plan (1 file)
- Formative + reflection (1 file)
- Assessment suite (1 file)